Anita Rao Mysore

Value of conscientiousness

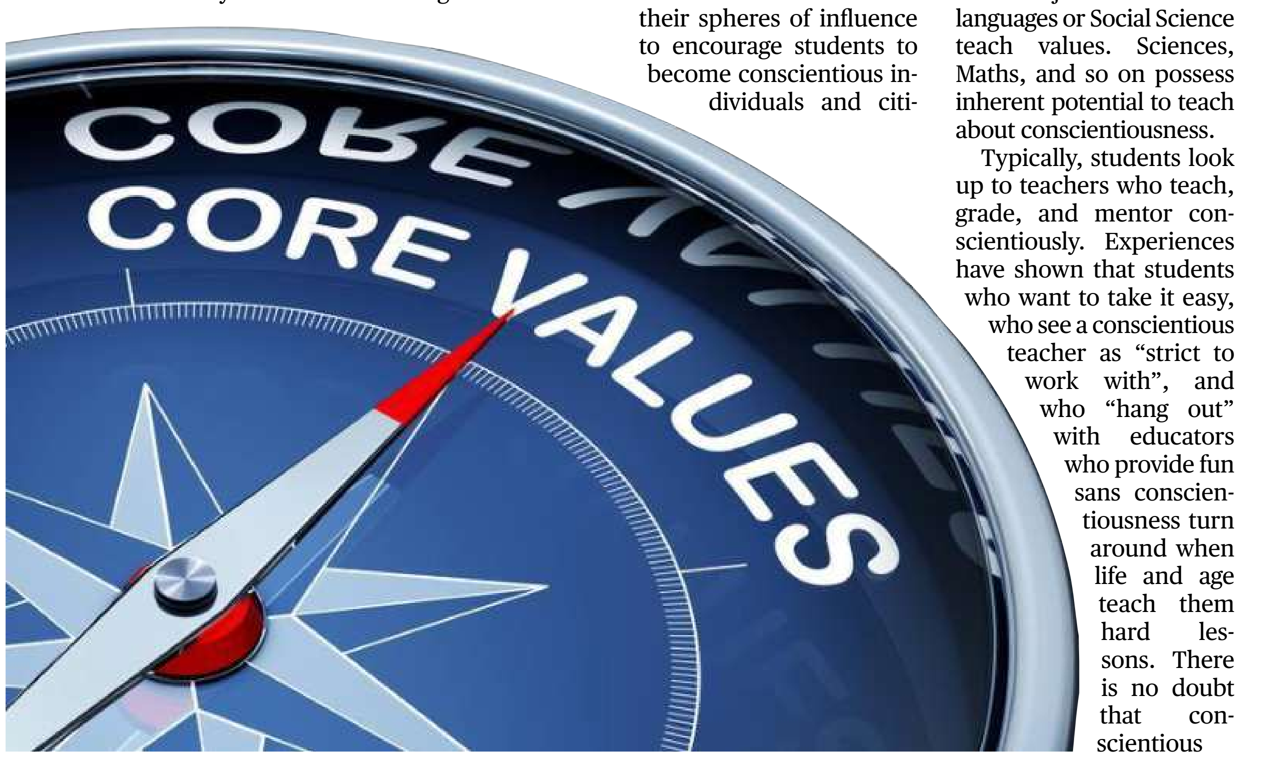
What educators and students can learn from the International Day of Conscience (April 5)

On July 25, 2019, the United Nations passed a resolution to declare April 5 as the International Day of Conscience. The rationale for "promoting the culture of peace with love and conscience," according to the UN, is: "Conscious of the need for the creation of conditions of stability and well-being and peaceful and friendly relations based on respect for human rights and fundamental freedoms for all without distinction as to race, sex, language or religion."

That the world craves for peace was evident in the recent viral videos and global following of the Walk for Peace by the Dhammacetiya monks and their dog, Aloka, in the U.S. Juxtaposed, however, are video games, media, and social media, comprising violence in different formats, in different genres, and spread across local to international locations geographically. In contemporary education, the unethical use of generative Artificial Intelligence (AI) in research and assessment in higher education and PK-12 schools respectively begs the need for conscientious individuals. When violent and/or dishonest means are normalised and begin to replace basic human values, it becomes imperative to make the International Day of Conscience more relevant and deliberate than ever in educational institutions.

For educators

What can educators do in their spheres of influence to encourage students to become conscientious individuals and citizens? While the cliched phrase in education states that "Values are to be caught, not just taught", there is value in both. Educators can teach students about conscientiousness via the three hats they wear: the unique positions that they hold, their subjects, and teaching courses such as value education, service-learning, global perspectives, life skills, and the like. Subjects other than languages or Social Science teach values. Sciences, Maths, and so on possess inherent potential to teach about conscientiousness.



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Typically, students look up to teachers who teach, grade, and mentor conscientiously. Experiences have shown that students who want to take it easy, who see a conscientious teacher as "strict to work with", and who "hang out" with educators who provide fun sans conscientiousness turn around when life and age teach them hard lessons. There is no doubt that conscientious teachers leave an indelible mark on the minds of their students.

For students

What can students do in their spheres of influence to become conscientious individuals? A student's ability to recognise and respond to the needs of those in the margins is a hallmark of conscientiousness. The act of giving is not confined to monetary donations; it extends to skill and time that students can give in service to society. Conscientiously using the planet's resources and living in harmony with nature are crucial today. Peace and partnerships are vital to a smoothly functioning society. Engaging with civility in "difficult dialogues" can foster peaceful relations and understanding.

Everyday life happens in the context of language, social and emotional interactions and is necessary even for professional settings. The lesson from the marshmallow experiment can be extrapolated to leading a conscientious life. In the 1960-70s, Walter Mischel, a professor of psychology at the U.S.' Stanford University, studied how the self-control skills of four year olds could relate to future success. Children were offered a choice of "one marshmallow now or two marshmallows if they waited for 15 minutes". Follow-up studies found that children who had waited for the second marshmallow by exercising self-control had obtained higher scores, had fewer health issues, and were better in social skills and stress response.

Taking short cuts seldom leads to permanent solutions and success! Conflict resolution and community engagement ought to go beyond requisites for credits, college admissions, and job CVs.

The art and science of self-reflection is a must for educators, students, and everyone. Self-reflection can make a person deeply conscientious. Journalling should not become a mere check in the checklist for "Becoming a More Productive Person." Reflection and journalling must pave the way for greater empathy and conscientiousness.

One day of observing the International Day of Conscience is not a panacea. Conscientiousness, when deeply rooted in individuals, can help transform the collective. This needs application of critical thinking and deliberate action. Conscientious families, neighbourhoods, communities, academics, businesses, and other social institutions are the bases for a healthy world. Educational institutions can use the International Day of Conscience as a springboard to recommit to being more and more conscientious, and to spread the word and deed in the community.

The writer is a retired associate professor, Education and Psychological Sciences, Christ University, Bengaluru.

Make the right choice

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OFF THE EDGE

Nandini Raman

Build your profile

Uncertain about your career options? Low on self-confidence? This column may help\

I am in Class 10. I am interested in MUN and AIPPM and am good at debates and public speaking. I wish to work for the UN, but don't know what courses or what degrees to do. I am not keen to write the UPSC. How can I work towards my goal? Isabel

Dear Isabel,

The UN and its agencies hire for a variety of roles across domains. Roles involving diplomacy, international relations, public policy and Law require strong writing, negotiation, and analytical skills. Those in development, social work, and public administration focus on projects with NGOs, community engagement, advocacy, and specific campaigns. They also need technical specialists across health, finance, statistics and IT. However, subject expertise and project experience are a must.

Since your interests are in MUN, debates, and public speaking, the Humanities or Social Sciences seem like a natural fit. Choose a course that excites you academically. Consider courses such as B.A. Political Science, International Relations, Economics and International Relations, Development Studies, Social Work, and so on. The five-year Integrated Law is another option.

Strengthen your skills, take on leadership roles, and get involved in social initiatives. Gain experience through internships or volunteering with NGOs in areas like human rights, environment, and development, and explore UN youth programmes. Learning a foreign language can also boost your prospects.

I did B.Sc. Cardiac Technology, and am interested in pursuing an MBA with any specialisation. Will my background affect selection or placements? Prateeksha

Dear Prateeksha,

If you want to leverage your B.Sc., choose an MBA in Healthcare or Hospital Management, as it will be the easiest for placements and career growth. Most MBA programmes in India accept graduates from any discipline, including paramedics and science. Admission primarily depends on entrance exams (CAT, XAT, GMAT, MAT, CMAT, and so on), academic performance (marks, consistency), work experience (optional but adds weight) and statement of purpose and interview.

Other industries will need specialised knowledge, analytical skills in operations, analytics, and consulting. So, attend workshops or short courses in general management or business analytics and explore internships and projects to understand the application of the MBA in a real setting. Network with alumni of MBA programmes from healthcare or science backgrounds and build LinkedIn connections in healthcare management and hospital operations.

I am in the fourth year of B.A. LLB. My college does not organise any events. How do I grow without support from my institution? Aarchi

Dear Aarchi,

Leverage online resources and use the Internet as your campus. Look for courses on Constitutional Law, International Law, Intellectual Property, Human Rights and Corporate Law on Coursera, edX, or NPTEL. Podcasts on channels such as LawSimplified or LegalEdge offer lectures, debates and analysis of judgments. Go through legal blogs to understand current trends. Read up on research papers in Law on SSRN, JSTOR, or HeinOnline. Try remote participation in online moot courts and debating competitions (for example, Manupatra Online Moot or VMUU Moot). Participate in online competitions hosted by the Bar Council or law firms. Join online groups for law students or young lawyers to build your peer and mentor network. Reach out to alumni from your college or law school for mentorship. Volunteer in legal aid cells, NGOs, or public policy forums to get practical exposure and build connections. Look for internships at law firms, corporate legal departments, NGOs, and legal research organisations. Write a paper on a legal topic and submit it to journals and freelance on platforms like Upwork or Lawctopus for legal writing and research projects

I finished B.Tech. Mechanical Engineering in 2022. I wasted two years not knowing what to do, and have just joined an auto manufacturing company as a trainee. My interests are more in cognitive work, innovation and AI. I am not interested in management, IT, or a regular 9-5 job. I am prepared to study further. What should I do next? Sumen

Dear Sumen,

You are clear about your interests in cognitive, innovative, AI-focused roles. This suggests that you are probably better suited for R&D, AI applications, interdisciplinary engineering, or design-thinking roles. Some potential domains to pivot into where mechanical, AI and cognitive work intersect would be robotics or automation, computational mechanics or simulation, AI & machine learning in Engineering, mechatronics or smart systems, product innovation and design, energy or sustainable tech or interdisciplinary research fields.

Consider a M.Tech or M.S. Mechanical Engineering or AI or Robotics either in India or abroad. Another option could be a Postgraduate Diploma or Certification in Robotics or Mechatronics. Simulation and optimisation courses could land you in industry roles in R&D or innovation labs. An interdisciplinary programme in Cognitive Computing and Robotics or AI and Mechanical Design will help you move from traditional mechanical to future-oriented cognitive roles.

Practical experience is crucial. Build a portfolio with projects in predictive maintenance for mechanical systems using AI, simulation-based design optimisation, robotics prototypes and so on. Participate in competitions of hackathons. Seek a research internship at IITs, DRDO, ISRO, automotive R&D labs in AI, smart systems, or optimisation in mechanical engineering.

Disclaimer: This column is merely a guiding voice and provides advice and suggestions on education and careers.

The writer is a practising counsellor and a trainer. Send your questions to eduplus.thehindu@gmail.com with the subject line Off the Edge

Sakhhi Chhabra

Beyond the brain drain

Why a nation of world-class minds fails to retain its own talent

FREEPIK

A few months ago, a friend expressed a concern that now feels almost universal among Indian parents. Despite the country's expanding higher-education map, she struggled to find a college that offered rigorous, practice-oriented learning for her son. Her search had already drifted toward universities abroad.

This led me to ask: What propels so many youngsters to look beyond their own borders? Is it the allure of a privileged global tag? Or does our higher education system fail to inspire confidence and offer growth opportunities? This is the paradox of modern India: a country that produces world-class talent but struggles to retain it.

As the Ministry of External Affairs reports, the number of Indian students going abroad has surged from 1.3 million in 2023 to more than 1.8 million in 2025. Data from the Reserve Bank of India's quarterly balance of payments further shows that this outward push has contributed to a $6-billion deficit. For a nation that proclaims itself an innovation hub, the irony is unmistakable: our finest minds continue to enrich foreign economies, laboratories, and knowledge systems.

Aspirational and economical

At one end are elite institutions with limited seats that push top scorers to look abroad; at the other, few Indian colleges offer jobs that repay tuition within a year, making overseas education both aspirational and economically sensible. Students often earn far more abroad for comparable roles, and many countries heighten this appeal with post-study work visas and pathways to long-term residency that add substantial future value. Layered onto this are anxieties about outdated pedagogy, the friction of urban living, and the desire for cleaner, freer, more cosmopolitan environments.

For some, foreign education is a release from India's relentless academic sorting mechanisms; for others, it becomes a chance to reinvent themselves, unburdened by the expectations and labels that shape their performance at home. The exodus reflects both India's structural gaps and a testament to the expanding horizons of its youth.

It is not enough to slow the outward flow of talent. What we need to do is to build an academic ecosystem that truly matches the ambitions of this generation. This requires moving beyond a handful of elite "islands of excellence" and cultivating a far broader landscape of schools and universities that offer credible, world-class environments. By 2047, India will need an education system that fosters confidence, curiosity, and creative risk-taking.

Change underway

While recent emphasis on strengthening foundational learning is a welcome step, a deeper contradiction is evident: India now has the world's second-largest network of international schools, growing from 884 in 2019 to 972 by January 2025.

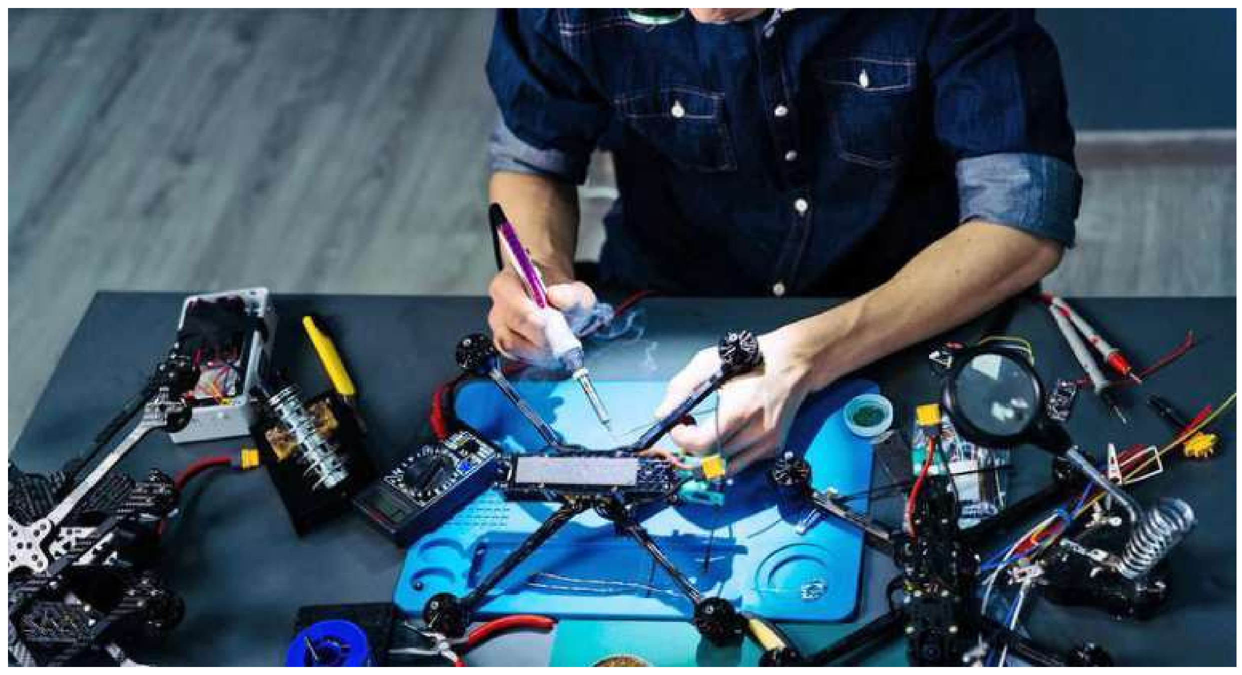
Even as primary and secondary education increasingly moves beyond rote learning, higher education remains ill-equipped to absorb these globally prepared students, nudging many to look abroad.

The contours of renewal are already visible. A new generation of Indian institutions is expanding research-driven, interdisciplinary undergraduate programmes that increasingly match global standards while private universities bring world-class pedagogy and international exposure to Indian campuses. Yet this promise will remain fragile unless structural barriers are addressed, most notably the need for transparent, merit-based admissions, improving research culture, building a connection between industry and academia, and decisive action against fraudulent colleges that erode trust in higher education.

These academic gaps intersect with broader anxieties due to rising unemployment and widening income disparities to stagnant wages in skilled sectors and cities that fall short of global standards of safety, hygiene, and liveability. An opportunity-rich society must offer not just quality education, but dignified employment, competitive pay, and workplaces free from discrimination. Educational renewal alone cannot anchor young talent; it must be matched by an economy that can absorb and reward it. Universities can ignite aspiration, but only a dynamic economic landscape can turn that aspiration into opportunity.

The priorities are clear: institutions that offer excellence at accessible costs; curricula that foster research, creativity, and critical thinking; and a labour market that rewards merit, supports entrepreneurship, and provides equitable pathways for all qualified graduates. Strengthening industry linkages, expanding recruitment pipelines, and improving urban infrastructure are central to rebuilding trust. When these forces align, the narrative can shift from escape to engagement.

The writer is an Assistant Professor at XLRI Delhi-NCR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Sketch, prototype, iterate

Why we need to reimagine Engineering pedagogy in India

Gautam Khanna

The year was 1903 and two bicycle mechanics from Dayton, Ohio, were about to change the world. They possessed no formal engineering degrees, no sophisticated laboratories, and no government grants. What they had instead was an unshakeable belief in learning by doing. Their journey to Kitty Hawk wasn't paved with textbooks, but with numerous prototypes and wind tunnels crafted from wooden boxes. When their first powered flight lasted just 12 seconds, it was more than a fleeting triumph; it was the culmination of years of hands-on experimentation.

Today, as India aspires to be a global innovation hub, we face the challenge of fostering innovators like the Wright brothers from the ground up. Our engineering education system produces around 1.5 million graduates annually, yet a massive hands-on skills gap persists. The issue is not a lack of intelligence, but a pedagogical failure to prioritise the "making" that defines true engineering.

Theory meets reality

For decades, India's engineering system has been optimised for scale over substance. Students understand abstract concepts but hesitate to design, build, or troubleshoot in the real world. This isn't just a failure of skill acquisition; it is a failure of pedagogy. We have historically treated "making" as an optional extracurricular activity rather than the core of the learning process. Unlike foundational innovators like Alexander Graham Bell, whose first telephone was a messy, crackling prototype of trial and error, our curriculum remains dense with theory and examinations that discourage the very experimentation required for true innovation.

This is not an argument against theory. Rigorous fundamentals matter deeply. But theory becomes transformative only when it is applied, because prototyping turns abstract knowledge into understanding. A student who designs a circuit and watches it fail learns more about current, resistance, and tolerance than one who solves 10 problem sets.

While established fields like medicine and law have centuries of history, Indian engineering is a recent phenomenon. India cannot boast of innovations to match the scale of students graduating every year. This is because the engineering academic system not only lacks the iterate-improvise-innovate prototyping methodology but also strongly discourages any deviation from established curricular practices, which are entirely incompatible with experimentation and innovation.

Across India, we're seeing glimpses of a better way to learn. Forward-thinking colleges are replacing step-by-step lab practicals with open-ended, multidisciplinary labs where faculty act as mentors rather than lecturers. The goal now is to move beyond these isolated success stories and make experiential learning pervasive across engineering education.

What can be done

Part of the answer lies in integration, not in addition. Prototyping cannot remain at the margins as a hobby club or optional course. It must be woven into the core curriculum, with policy and institutional support to sustain it. Students should build before they specialise, and assessment systems must value process: how students think, test, and refine, not just final answers.

Equally important is faculty enablement. Many teachers were educated in theory-heavy systems. Expecting them to suddenly shift pedagogical styles without support is unrealistic. Structured investment in faculty development, industry engagement, and shared infrastructure can lower this barrier.

A deeper cultural shift is also required. Critically, this shift is not about glorifying entrepreneurship. Not every student needs to become an inventor or founder. But every engineer, whether designing bridges, writing software, or managing systems, benefits from having once taken an idea from sketch to prototype to iteration. It builds respect for constraints, empathy for users, and humility before complexity.

The Wright brothers' first flight was wobbly, barely airborne, and covered less ground than the wingspan of a modern airliner. But it proved that flight was possible, and everything that followed was iteration. Somewhere in India today, a student is watching their first prototype sputter to life on a workbench. It won't be elegant. It probably won't work on the first try. But in that moment, they are learning the way engineers were always meant to learn. Our job is to make sure they don't have to do it despite their education, but because of it.

The writer is the CEO of the STEM Siksha Foundation.

First line of defence

Why defence R&D exposure should be part of Indian STEM education pathways

Sudip Das

According to the All India Survey on Higher Education (AISHE), India's higher education system has nearly 30 million students enrolled across roughly 50,000 higher education institutions. This scale underscores the potential of the Engineering pipeline. However, there is a gap between ambition and capability in advanced defence research and systems engineering. These two facts are rarely examined together within higher education and capability planning.

More theoretical

Across Engineering programmes, the constraint is not a lack of student interest. In fact, defence technologies draw sustained curiosity. Students are fluent in topics like missiles, radars, propulsion, cryptography, and autonomous systems. What they lack is not intent, but exposure. For most, defence research remains theoretical until late postgraduate study.

Defence systems are not mastered through short projects or compressed training modules. They demand systems thinking across hardware and software, tolerance for failure, security discipline, and the ability to operate under constrained, high risk conditions. These capabilities are not acquired through a final-year dissertation or a brief internship.

Interest in mission-oriented public research is visible early in an engineer's academic life. When engagement is postponed, that interest often dissipates. Students redirect towards domains that offer faster validation and clearer career signalling, driven more by incentive structures than ability or intent. This gap matters because the nature of India's defence engagement itself is changing.

Capability is increasingly framed not in terms of one-time procurement but as lifecycle responsibility, covering integration, maintenance, upgrades, and indigenous redesign across air, maritime, and aerospace systems. Such work extends over decades. It depends on deep domestic capacity in systems engineering, materials science, embedded software, signal processing, and reliability engineering.

Policy assessment of India's higher education system has identified critical execution gaps in academic programmes, research exposure, and national capabilities priorities. However, the persistence of this gap is not accidental. Defence research operates under legitimate security constraints, while universities function with rigid approval cycles. Risk aversion becomes structural. Security concerns often lead to avoidance rather than pedagogical design, leaving exposure deferred and poorly aligned with early-stage student learning. As a result, defence is presented as a specialisation rather than a foundational pathway. Exposure that arrives only in the final year functions as selection, not formation.

What can be done

However, policy intent has changed. Initial exposure, interdisciplinary learning, and applied problem-solving are prioritised in undergraduate education according to the National Education Policy 2020. The question is no longer permission. It is execution. There is a discrepancy between institutional practice and policy language. It takes administrative confidence, faculty incentives, and curriculum flexibility to turn intent into exposure. Defence-related electives that are credit-bearing and lab-related can be included in the undergraduate curriculum. Under specific disclosure guidelines, capstone projects can be co-designed with public research labs. Summer programmes can offer problem statements grounded in real platforms rather than abstract case studies detached from Indian operating conditions.

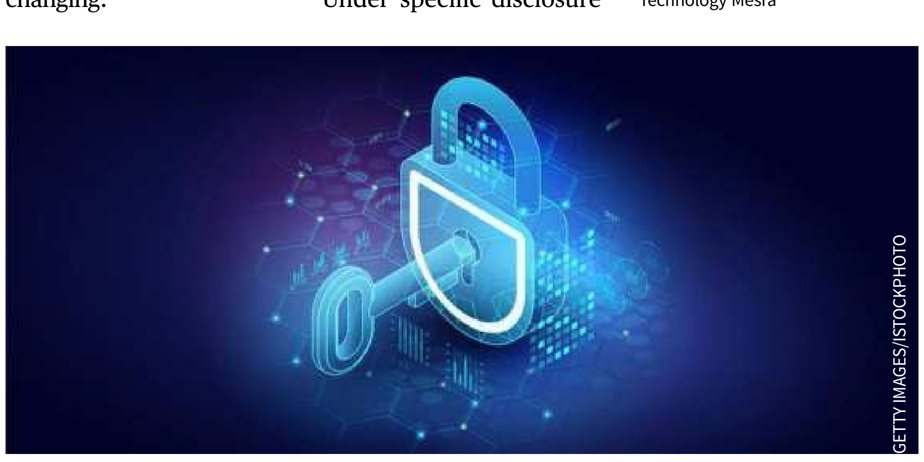
Faculty engagement is equally critical. Without instructors who understand defence systems and constraints, exposure risks become superficial. The lack of organised channels to convert research experience into classroom teaching, project schedules, and access restrictions all contribute to the uneven interaction of faculty with strategic research ecosystems. Reform requires faculty development, collaborative supervision, and common infrastructure; these are not optional.

Discussions around self-reliance often reduce themselves to procurement targets and manufacturing percentages. Intellectual autonomy is harder to measure, but more decisive. When core algorithms, materials research, propulsion logic, and system architectures are developed elsewhere, dependence follows quietly.

A guidance algorithm refined in one's 20s may still be operational decades later. Advancements in defence research build the student's capabilities in embedded systems, artificial intelligence, materials science, thermal engineering, and hardware-centric cybersecurity that continue to matter across sectors.

The question, then, is not whether India can afford early defence exposure in STEM education. It is whether it can afford a pipeline that consistently engages its engineers only after formative years have passed.

The writer is a Professor in the Department of Space Engineering and Rocketry at Birla Institute of Technology Mesra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THINK
Aruna Sankaranarayanan

Walls up?

How to deal with defensiveness

Whether it's a colleague, friend or parent, dealing with defensiveness can be challenging. When you raise a concern, the person is not open or ready to accept that there is a problem in the first place. Defensiveness can manifest in a variety of contexts, ranging from prosaic activities to more poignant ones. A coworker may repeatedly leave his unwashed coffee mug in the conference room. When you bring it up, he brushes it off as a one-time issue. A friend is habitually late. When you text her to be on time for a lunch date, she says she can't make it. You tell your parents that you plan to move out by the end of the month. Your father changes the topic asking whether you watched the cricket match last night.

In an article in *Psyche*, Adar Cohen, a mediator, and Nick Wignall, a psychologist, unpack the psychology of defensiveness and provide strategies for having more fruitful conversations when we encounter it. According to the authors, defensiveness is a "coping strategy" that people engage in when they're unable to deal with the "painful feelings" that arise when they're criticised; so, they throw the "blame back on the criticiser".

In other words, defensiveness occurs when a person is unable to handle their own anxiety or discomfort. When you try to parse a person's defensiveness through this lens, you are more likely to exhibit compassion and respond in a manner that's less threatening to them.

Further, the authors aver that it's pointless to tell someone to stop being defensive. Because defensiveness is a feeling, it's impossible for the person to voluntarily curb it. In fact, telling them to do so will only increase their anxiety and frustration and make them more "resistant to your suggestions". You also want to be careful not to trigger a never-ending cycle of defensiveness, wherein you become defensive in response to their reaction.

How you react

As you cannot change how the other person feels, your best bet is to try and tame your own emotions. The authors recommend practising a technique called AVA, wherein you "acknowledge, validate, act". Suppose you bring up the topic of moving out with your dad, and he shifts to another topic. Your first instinct is to get irritated and yell at him. However, you don't want this conversation to take an unpleasant turn right from the very start.

So, the first step is to acknowledge your feelings to yourself. You are irritated and frustrated that your dad is refusing to have this conversation with you. Then validate these feelings by giving yourself permission to feel them. Tell yourself that it is okay to have these feelings as a young adult who craves autonomy. Finally, act on your values. You love your dad and do not want to jeopardise your relationship with him.

When you broach the subject of moving out, you may start with something like, "I know this is hard for you to hear. I would like to move out. That does not mean that I will stop being close to you." While your dad is unlikely to come around immediately, positioning your desire in this way is more likely to lead to a conversation.

Your dad may say, "As if you lack anything here..." Instead of countering it with defensiveness, try to exhibit curiosity by asking, "What worries you about my moving out?" Be watchful that your tone is sincere, not sarcastic. If you anticipate another person's defensiveness, you also need to pick the right time and place for the conversation to minimise their discomfort.

While the AVA method may not entirely alter another person's defensiveness, it allows you to gain some distance from your emotions. As a result, you don't get carried away by them, and are able to act in accordance with your values.

The writer is visiting faculty at the School of Education, Azim Premji University, Bengaluru, and the co-author of Bee-Witched.

From clay to code

What students need to know about digital manufacturing

Patrick J. Smith

Manufacturing is a defining characteristic of humanity, involving the creation of objects to meet anticipated needs. As societies settled and farming spread, it evolved from a simple skill into a specialised craft, producing goods at scale for trade. The invention of the potter's wheel, for example, increased pottery production tenfold: basic pots became widely available while elaborate ones remained prestige items.

What it is

Now, we are witnessing the next major leap in this journey: digital manufacturing. Unlike traditional manufacturing, which relies heavily on human labour and mechanical tools, digital manufacturing integrates data, automation, and AI to design, produce, and optimise products in real-time. Digital manufacturing also allows companies to manage operations across multiple sites and countries through enterprise-wide data platforms, improving efficiency, predictive maintenance, supply-chain management, and sustainability reporting.

Previous industrial revolutions focused on mass production, improved communication, and the rise of information technology. Industry 4.0 builds on this by enabling real-time decision-making, higher productivity, and flexibility in how products are made and delivered. Examples include automation such as driverless vehicles and precision agriculture powered by machine learning, robotics, and advanced analytics.

New digital tools are also creating innovative business models, letting manufacturers offer services alongside products, such as predictive maintenance or usage-based billing, benefiting both companies and customers. AI powers these systems, helping forecast demand, optimise energy use, and detect machine anomalies beyond human capability.

India is investing heavily in smart factories, automation and industry 4.0 initiatives to modernise sectors such as automotive, electronics, pharmaceuticals, and textiles. The growing adoption of robotics, IoT, and AI in production are creating a demand for a workforce skilled in digital tools and data-driven decision-making.

For students, this shift means that acquiring knowledge in areas such as automation, 3D printing and data analytics, and sustainable manufacturing, is not only globally relevant but also tied directly to India's industrial growth. By developing these skills, they can shape modern factories, contribute to innovation, and access exciting, high-growth career opportunities.

Necessary skills

To meet this demand, students need to develop a deep understanding of highly productive fabrication processes and work with advanced software systems capable of self-analysis and complex problem-solving. Alongside technical skills, they must build the critical thinking skills and intellectual curiosity required to challenge existing practices and drive innovation. A strong emphasis on high-productivity manufacturing and understanding how digital control systems and machine-based decision-making can automate downtime and routine operations to deliver significant gains in efficiency and output is crucial.

Thus, what is required is a combination of technical, analytical, and practical skills. Students must learn to work with automation and robotics, digital design and 3D modelling, and data analysis to optimise production and support real-time decision-making. Experience in managing complex operations across multiple sites and applying sustainable manufacturing practices to reduce waste and energy use is another aspect. Hands-on projects and industry-focused learning will help develop problem-solving, teamwork, and project-management skills.

These competencies prepare graduates to innovate, lead, and shape the factories of the future in a rapidly evolving, data-driven manufacturing landscape. Digital manufacturing is not just a technological shift but a career revolution. Roles are emerging in automation, robotics, data analytics, digital design, and innovation management. Embracing these opportunities will place young professionals at the forefront of a global and national industrial transformation.

The writer is Senior Lecturer in Manufacturing Technology, University of Sheffield, the U.K.